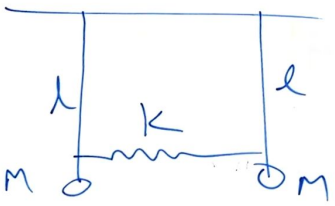
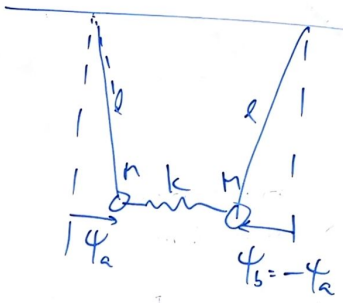
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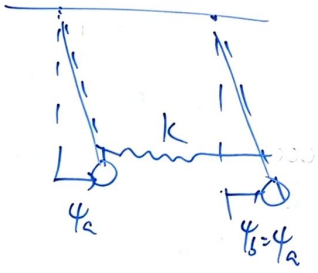
# Coupled pendulum oscillations, Beats – Waves Lec 24




**Mode 2** : $\psi_b = -\psi_a$




**Mode 1** : $\psi_b = \psi_a$




① $\omega_1^2 = g/l$

② $\omega_2^2 = \frac{g}{l} + \frac{2k}{M}$

$\omega_2^2 - \omega_1^2 = \frac{2k}{M}$

$\psi_a = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$ (3)

$\psi_b = A_1 \cos(\omega_1 t + \varphi_1) - A_2 \cos(\omega_2 t + \varphi_2)$ (4)

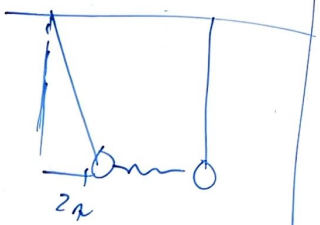
If (5) $\psi_a(0) = 2A$, $\psi_b(0) = 0$

$\dot{\psi}_a(0) = 0$, $\dot{\psi}_b(0) = 0$

(6) $\begin{cases} A_1 = A_2 = A, \\ \varphi_1 = \varphi_2 = 0 \end{cases}$

(7) $\psi_a(t) = A \cos\omega_1 t + A \cos\omega_2 t$

(8) $\psi_b(t) = A \cos\omega_1 t - A \cos\omega_2 t$




$l \approx 1$, $g = 9.8 \frac{M}{s^2}$

$M \approx 1\,kg$, $k = \frac{1}{2} N/M$

$\omega_1 \approx \sqrt{9.8} \approx 3.1$

$\omega_2 = \sqrt{9.8 + 1}$

$\approx 3.3$

$\bar{\omega} =$

$\psi_a$

## Beats

$\bar{\omega} = \omega_{av} \equiv \frac{\omega_1 + \omega_2}{2}$ = Average of two freqs

$\omega_{mod} \equiv \frac{\omega_2 - \omega_1}{2}$ = Modulation freq

$|\omega_{mod}| \ll \bar{\omega}$

$\omega_2 = \bar{\omega} + \omega_{mod}$

$\omega_1 = \bar{\omega} - \omega_{mod}$

$\psi_a = A\left[\cos(\bar{\omega} - \omega_{mod})t + \cos(\bar{\omega} + \omega_{mod})t\right]$

$= A\left[\cos(\bar{\omega}t - \omega_{mod}t) + \cos(\bar{\omega}t + \omega_{mod}t)\right]$

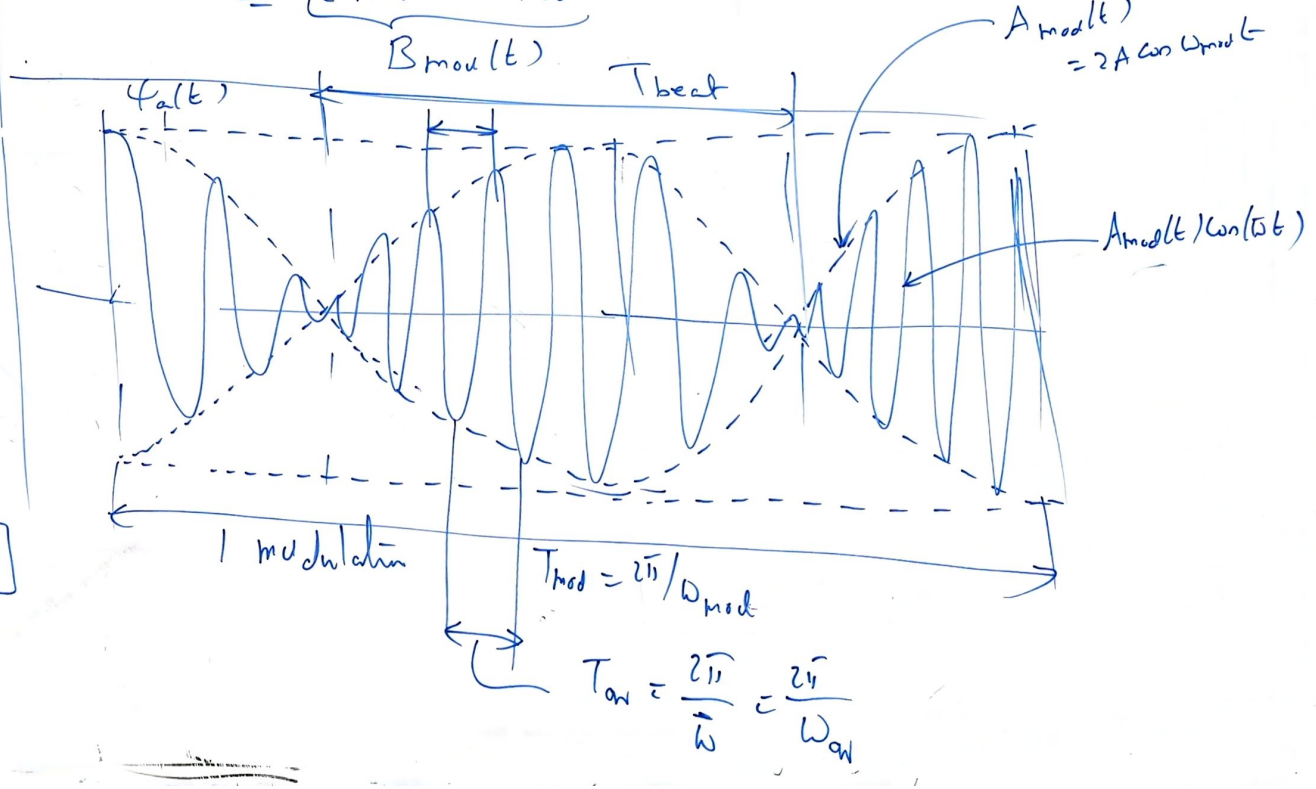
$= \underbrace{2A\cos\omega_{mod}t}\ \cos\bar{\omega}t$

$A_{mod}(t)$ → slowly varying

$\psi_b(t) = A\left[\cos(\bar{\omega}t - \omega_{mod}t) - \cos(\bar{\omega}t + \omega_{mod}t)\right]$

$= \underbrace{2A\sin\omega_{mod}t}\ \sin\bar{\omega}t$

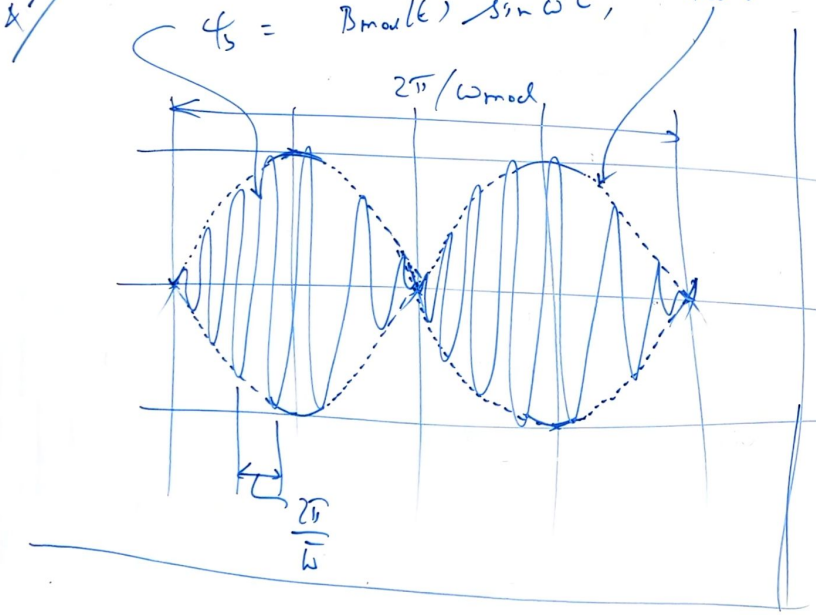
$B_{mod}(t)$



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# Coupled pendulum oscillations, Beats – Waves Lec 24

$\psi_s = B_{mod}(t) \sin \bar{\omega} t$, $B_{mod}(t) = 2A \sin \omega_{mod} t$




What we hear is $\propto$ Amplitude$^2$

$= A_{mod}^2(t)$ !

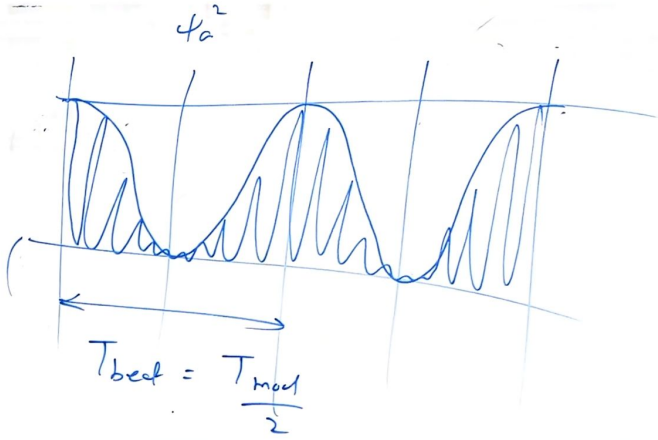
$= 4A^2 \cos^2(\omega_{mod} t)$

$= 2A^2 [1 + \cos 2\omega_{mod} t]$

$= 2A^2 [1 + \cos(\omega_2 - \omega_1) t]$

$\omega_{beat} = |\omega_2 - \omega_1| = 2\omega_{mod}$

$T_{beat} = \frac{2\pi}{\omega_{beat}} = \frac{2\pi}{|\omega_2 - \omega_1|} = \frac{1}{2} T_{mod}$




3.3